

Table 1 Tools for designing DNA sequence libraries and for related sequence-design problems. Rows appear in the same order as the columns of Table 2. The final two rows group tools that address problems adjacent to library design; their members are named and cited individually. Tools examined August 2026.

Tool	Purpose	Key features	Output	Availability	Ref.
DNA Chisel	Optimise a single sequence against declared constraints and objectives	Hard constraints and scored objectives combined in one problem; codon optimisation and avoidance of restriction sites, homopolymers and extreme GC	One optimised sequence with a constraint-satisfaction report	Python package (PyPI); web application	[1]
MPRA Design Tools	Design MPRA libraries and estimate statistical power	Barcode assignment with error-correcting sets; power analysis over effect size and sequencing depth	Barcoded constructs and statistical power estimates	R package (GitHub); Shiny application	[2]
MPRAnator	Design MPRA libraries of motif arrangements and sequence variants	Combinatorial placement of motifs; randomly sampled and combinatorial substitution sets; scrambled and reverse-complement controls	Library of sequences (FASTA)	Web service	[3]
Mutation Maker	Design mutagenic primers for laboratory construction of variant libraries	Three workflows: site-scanning saturation mutagenesis, multi-site directed mutagenesis, and PCR-based gene synthesis; degenerate codons and primer mixing ratios	Mutagenic primer pairs, or gene fragments with mixing ratios	Web application (Docker); REST API	[4]
Oligopool Calculator	Design library infrastructure and analyse sequencing readout	Constraint-aware barcodes, primers and spacers; degenerate compression of a supplied variant set; off-target screening against a host genome	Synthesis-ready library; variant counts from sequencing reads	Python package (PyPI); command line	[5]
PoolParty	Design DNA sequence libraries of any type	Libraries specified as a directed acyclic graph of operations; saturation mutagenesis, randomly sampled variants and pairwise and higher-order variants can be mixed in one library; per-variant design cards	Library of sequences, each with a design card (CSV, TSV, FASTA, JSONL)	Python package (PyPI)	This work
tangermeme	Probe cis-regulatory logic with deep learning models	Saturation mutagenesis of input sequences; insertion and removal of motifs across sets of sequences	Perturbed sequences and model predictions	Python package (PyPI)	[6]
ValiAnT	Design and annotate libraries for saturation genome editing and cDNA deep mutational scanning	Saturation mutagenesis from genomic coordinates; several mutation types per target region; transcript-aware, including codons split across exons	Per-region libraries with metadata tables and VCF carrying variant consequence annotation	Command-line tool (source, Docker)	[7]
General-purpose toolkits (<i>Biopython</i> , <i>pydna</i> , <i>SeqPro</i>)	Manipulate sequences, simulate cloning, prepare arrays for modelling	Parsing, transformation and file-format support; no library abstraction	Transformed sequences	Python packages (PyPI)	[8–10]
Single-sequence design tools (<i>CodonGenie</i> , <i>ledidi</i>)	Choose codons, or edit one sequence toward a model's objective	Degenerate codons covering a target residue set; gradient-based editing guided by a predictive model	A codon set, or one edited sequence	Web service; Python package	[11, 12]

Table 2 Capabilities of tools for designing DNA sequence libraries. ● denotes a capability the tool provides; ◐ a capability provided with a stated restriction; ○ a capability the tool does not provide. A capability counts only where the tool supplies an operation, parameter or mode for it, and not where a user reconstructs it by composing primitives the tool provides for other purposes. Tools assessed August 2026 against their documentation, source repositories and, where installable, executed code. Operational definitions of every row are given in Additional file 1.

	DNA Chisel	MPRA Design Tools	MPRAnator	Mutation Maker	Oligopool Calculator	PoolParty	tangerneme	ValiAnT
<i>Library Specification</i>								
Library object	○	◐	○	◐	◐	●	◐	○
Composable operations	◐	○	◐	○	●	●	◐	○
Mixed variant types in one library	○	●	●	◐	○	●	○	●
<i>Variant Generation</i>								
Saturation mutagenesis	○	○	○	○	◐	●	●	●
Randomly sampled variants	●	◐	●	○	○	●	○	○
Pairwise and higher-order variants	●	○	●	●	◐	●	◐	○
Model-guided variants	●	○	○	○	○	◐	●	○
<i>Variant Types</i>								
Codon / amino-acid substitutions	●	○	○	◐	○	●	○	●
Insertions and deletions	○	●	◐	○	○	●	●	●
Combinatorial multi-motif placement	○	○	●	○	○	●	●	○
Recombination	○	○	○	○	○	●	○	○
Shuffling	○	○	◐	○	○	●	●	○
<i>Physical Construction</i>								
Synthesis-constraint checking	●	◐	◐	○	●	◐	○	◐
Constraint-based optimisation	●	◐	◐	●	●	○	○	○
Primer design	◐	○	○	●	●	○	○	○
<i>Metadata and Inspection</i>								
Per-sequence design cards	●	●	◐	●	●	●	○	●
Automatic naming	○	◐	●	●	◐	●	◐	●
Sequence styling	○	○	◐	○	○	●	○	○
<i>Genomic Integration</i>								
Genome coordinates	○	◐	●	○	○	◐	◐	●
Transcript / annotation aware	○	○	○	○	○	○	○	●

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